Iterative Agent-Team Updating for Evidence-Grounded Peer Review: Immutable Track 2 Execution and a 10-Paper Hosted Audit

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Abstract

We present an end-to-end Track 2 workflow with two bounded loops: paper-time review refinement over immutable evidence, and offline, development-gated updating of the review agent team's operating state. The persistent loop iteratively selects train-only changes to rubric weights, score calibration, and role-separated retrieval memory; it does not fine-tune hosted model weights. The fast-v1 paper-time path uses five base calls—two extractors, two reviewers, and one chair—plus a conditional two-call Author–Chair refinement. On a fixed ten-paper public benchmark, it produced 10/10 schema- and evidence-valid reviews in 631.9 seconds with 54 calls and no retry, failure, or timeout. Its 2.93 MB of serialized requests is an 83.24% reduction from a 17.50 MB counterfactual legacy serialization, but misses the registered 85% target. A public-paper case study provides diagnostic process evidence; absent human ratings, neither loop establishes reviewer-quality improvement.

1. Scope and Contract

Track 2 boundary. Here, Track 2 denotes the review operation itself, not a post-hoc wrapper around another pipeline. Its input is a final paper in PDF or Markdown and, optionally, pre-existing evidence such as result files, tables, logs, and code-version records. It neither edits the paper nor invents experiments. Claims that cannot be checked from the frozen bundle must be marked evidence-insufficient.

The implementation copies each input into a per-paper workspace, records its SHA-256, fixes the output path, and writes review-agent.md. PDF text extraction is also versioned and hashed. The resulting bundle digest binds the paper, evidence inventory, instructions, extractor, and output contract. Only these fixed inputs are visible to the review run. This realizes the preparation sequence specified in plan.md and prevents evidence from being silently added after review.

Output contract. The chair returns Summary, Strengths, Weaknesses, Questions for the Authors, Contribution, Overall Recommendation, Confidence, and Ethics and Limitations. The machine record additionally carries integer Soundness, Presentation, Significance, and Originality scores on 1–4 scales, Overall Recommendation on 1–6, and Confidence on 1–5. Python validation rejects missing fields, out-of-range values, malformed lists, unknown evidence identifiers, and unregistered contradictions. The Markdown renderer preserves the requested human-facing sections.

The system is restricted to public papers and author-authorized material. It must not process confidential submissions or private review discussion, nor submit generated text as an official review. These boundaries follow ICML's reviewer, LLM-use, and peer-review ethics policies (International Conference on Machine Learning, 2026d;c;b).

2. Fast Pipeline and Two Loops

Five-call base path. fast-v1 replaces the legacy 19-section extraction and 14-agent fan-out with two complementary extraction calls. Each returns typed claims, limitations, equations, experiment facts, and internal contradictions with stable evidence IDs. Two independent reviewers receive the compact evidence registry and specialize in technical validity and empirical/reproducibility checks. A chair consolidates their evidence-linked findings into the strict review form. In a ten-paper batch, all papers complete the five-call base path before refinement selection. At most two papers then receive an author-style challenge and a second chair call. This base-first scheduler bounds cost

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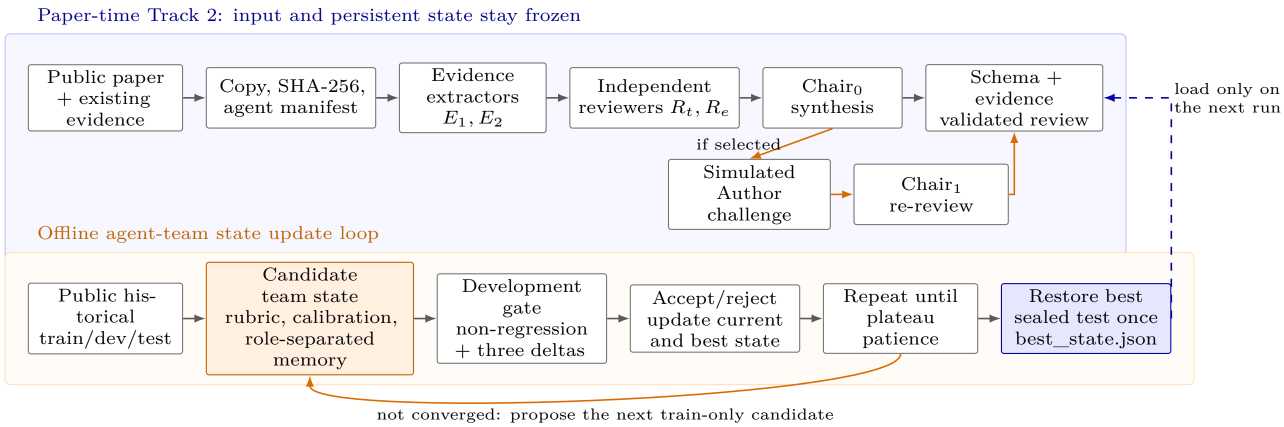

Figure 1. End-to-end workflow and two separated loops. Paper-time refinement stress-tests and may revise a review without changing its immutable bundle or persistent state. The offline loop iteratively updates rubric, calibration, and role-separated retrieval memory, admits candidates through a development gate, restores the best state, and supplies only the next Track 2 run. This is an operating-state update: hosted weights, prompts, and team roster remain fixed.

and prevents early papers from consuming the refinement budget.

The review-time state contains rubric weights, score calibration, and at most eight paper-agnostic memory items. Retrieval excludes the current forum and passes generalized checklists rather than stored reviews or rebuttals. Human reviews, rebuttals, and decisions are absent from paper-time requests. The public 20-case ReviewBench–ReviewRebuttal seed (ReviewBench Dataset Maintainers, 2026; ReviewRebuttal Dataset Maintainers, 2026) is split by forum into 16 train, two development, and two sealed-test cases. It is used only for a deterministic integrity/non-regression smoke test; it does not establish reviewer quality.

Iterative agent-team update. Figure 1 makes the persistent update loop explicit. Each offline iteration proposes a train-only, immutable LearningState candidate that updates rubric weights, calibration scale/bias, and role-separated reviewer/author memory. Development non-regression decides acceptance; quality, behavior, and state deltas drive plateau stopping; the best state is restored before one sealed-test evaluation. The registered seed run executed five iterations, selected iteration 1, and stopped after a three-step plateau. This is an orchestration-level team update, not gradient training, hosted-model fine-tuning, prompt rewriting, or roster adaptation. Separately, the paper-time simulated Author–Chair loop can revise only the current review over frozen evidence; it cannot teach later papers.

Backend isolation. The first-class hosted backend invokes Codex CLI 0.144.1 with an explicit gpt-5.6-sol model (OpenAI, 2026). Authentication is the local ChatGPT login rather than a repository API key. Each request runs in an ephemeral, read-only working directory with interactive and network tools disabled. A small environment allowlist excludes variables such as OPENAI_API_KEY. JSON Schema constrains model output; Python performs semantic validation and bounded retry. One global semaphore enforces concurrency four across papers, while a monotonic hard deadline terminates the whole process group. Atomic completion markers, content-addressed cache keys, and JSONL progress records support auditable resume without treating an invalid response as complete.

3. Evaluation Protocol

Frozen benchmark. The runtime audit uses the first ten entries of a public paper-only manifest. No human reviews, rebuttals, decisions, or private evidence are included. The set overlaps the historical split as eight train and two development forums, with no sealed-test forum, so it is a systems audit rather than a quality test. The manifest SHA-256 begins fdfdd199e694; the selected learned state digest begins fe1a1597e2e7. The registered configuration is four paper workers, global Codex concurrency four, two attempts per logical call, a 1,440-second soft deadline, a 1,800-second hard deadline, conditional author loop, and at most two refinements. The measured run is cold: cache hits are zero.

We validate (i) all JSON and rendered review schemas, (ii) every cited evidence ID against its paper registry, (iii) representative maximum request size relative to extracted paper text, (iv) deadline and call budget,

Table 1. Cold hosted audit on ten public papers.

Metric	Result
Schema/evidence-valid reviews	10/10
Wall time / throughput	631.9 s / 0.949 papers/min
Logical/actual calls	54 / 54
Retries / failures / timeouts	0 / 0 / 0
Request / output bytes	2,932,889 / 378,772
Linked findings / major–fatal	220 / 97
Max request-to-paper ratio	1.103
Deadline and call budget	Pass
Legacy request estimate	17,503,551 bytes, 309 calls
Request-byte reduction	83.24% (85% target: Fail)

Table 2. Concurrent per-call hosted latency in seconds. Refinement stages have two calls; durations are not additive wall time.

Stage	p50	p95
Evidence extraction	51.93	60.36
Consolidated review	38.00	42.10
Chair/final review	30.72	38.60
Author challenge	43.15	46.67

and (v) historical non-regression within tolerance 0.02. The quality-adjacent audit counts linked findings and contradictions, but does not label them correct.

Counterfactual baseline. For a like-for-like communication estimate, the validator replays the same papers, state, fast-v1 evidence, critiques, and final Markdown through the legacy request serializers without making model calls. This isolates request-shape overhead, but it is not a legacy end-to-end latency or quality measurement. The pre-registered acceptance target is at least 85% fewer request bytes than this serialization.

4. Results

Table 1 reports the run identified by the immutable input manifest, pipeline digest, state digest, model, and CLI version. Fifty calls served the base paths and four served the two selected refinements. All 220 finding/contradiction links resolved to registered evidence, including 97 items marked major or fatal by the generated records. This is a traceability count, not 97 independently verified defects. Byte counts are UTF-8 serialized backend payload and output sizes, not token usage, network transfer, or billing measurements.

The representative request ratio of 1.103 is below the registered ceiling of 3, and the run finished in 35.1% of the hard deadline. The counterfactual legacy serializer would issue 309 calls and 17.50 MB of requests. The observed 83.24% byte reduction is large but 1.76 percentage points below the 85% criterion; the overall acceptance audit therefore fails. Reducing shared schema and context repetition is the remaining measurable optimization.

Historical smoke. State build run 3842327ddf72639976d7 preserved development MAE at 0.7083 while Brier changed from 0.2863 to 0.2796. The one-time, two-forum sealed test reported MAE 0.3750 and Brier 0.1229. With only two development and two test forums and deterministic proxy reviews, these numbers verify state selection, role separation, and non-regression plumbing only.

5. Public-Paper Case Study

We separately ran the same Track 2 workflow on public paper 2602.00521v2; its PDF SHA-256 begins 6c29c6497761 and bundle digest begins bfacf0180851. Seven hosted calls completed in 220.2 seconds with no retry, failure, timeout, or cache hit, using 565,696 request bytes and 49,823 output bytes. The final scores were Soundness 1, Presentation 2, Significance 3, Originality 3, Overall Recommendation 2, and Confidence 4. Refinement retained these scores and continued to mark the review as needing refinement.

The final review identified a central latent-scale comparability problem in independently fitted human and LLM graded-response models; an equation denominator and undefined boundary cases; inconsistent threshold counts; gate inequalities and Phase-2 application mismatches; a prompt/JSON score-scale mismatch; and missing bootstrap unit and multiplicity specifications. These findings demonstrate that the evidence registry can support concrete, cross-section criticism. They were not adjudicated by the authors or blinded human reviewers, and some detailed table cross-checks were still missed. Accordingly, the case study is inspectable process evidence, not a precision, recall, or superiority result.

6. Limitations and Reproducibility

The benchmark is small, public-only, and contains revised paper text; RecSys is not represented in the historical seed. Hosted model behavior and service latency can drift. The byte baseline reuses fast-v1 intermediate outputs and therefore estimates legacy serialization rather than executing the legacy model pipeline. Evidence-ID validity establishes referential integrity, not factual correctness. No human annotation, inter-rater agreement, blinded comparison, review usefulness score, robustness suite, model ablation, or cost measurement was performed. Confidential-review suitability is explicitly out of scope. The case-study manifest omits its source URI, so its public provenance is not independently attested by that artifact and should be added before external release.

All reported values are generated artifacts rather than transcribed logs. The benchmark manifest, per-paper Track 2 bundles, progress records, reviews, run summary, acceptance audit, and human inspection sheet are under artifacts/benchmark_fast_v1/gpt-5.6-sol-cold. The case-study review is under artifacts/track2/2602.00521v2-loop2. The report builds with make -C report check. Formatting follows the ICML 2026 author style (International Conference on Machine Learning, 2026a); it does not imply submission or endorsement.

Conclusion. fast-v1 establishes a reproducible Track 2 path from immutable public input to evidence-linked structured review, while the dev-gated offline loop provides a versioned mechanism for repeatedly updating the agent team's operating state. The ten-paper run met schema, traceability, deadline, reliability, context-size, and historical non-regression checks. It did not meet the 85% request-byte reduction target, and neither loop yet supports a claim of improved reviewer quality.

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